

On the way home from a routine visit to the pediatrician, Sarah stops with her infant son at the local convenience store. As she is waiting in line, a man squeezes in closely behind them and sneezes. As Sarah walks home, she finds that her heart is racing, she feels nauseous, and she is terrified that her child will become severely ill.

Soon after, she becomes obsessed with disease and contamination. She begins to take extensive measures to "decontaminate" herself and her home, including washing her hands dozens of times daily. These cleaning rituals have the immediate effect of lessening Sarah's anxiety, so she soon spends most of her day performing them. She seeks treatment from a therapist, who instructs her to gradually expose herself to "contamination" but refrain from her cleaning rituals. The therapist also teaches Sarah how to recognize and modify her maladaptive thoughts about illness prevention. Within several weeks, Sarah finds that she can manage her thoughts about disease and contamination without resorting to ritualistic cleaning.

A researcher hypothesized that individuals with obsessive-compulsive disorder (OCD), such as Sarah, would show a more generalized fear response to safe stimuli compared to controls. Generalized conditioned fear responses occur when individuals react with fear to stimuli that resemble the original conditioned stimulus (eg, someone injured in an accident with a red truck fears all red vehicles). The researcher conditioned 100 individuals (50 with OCD and 50 controls) to associate a shock with a visual stimulus. In phase 1, participants viewed a computer screen that alternated between images of a small ring (1 cm) and a large ring (20 cm). The small ring and large ring were displayed for 1 second each, with a blank screen in between for 5 seconds. A mild electric shock was paired with the large ring, and no shock was paired with the small ring. This was repeated for 10 trials. In phase 2, shocks were discontinued, and three intermediate-sized rings were added to the series and presented in increasing size (ie, 1 cm, 5 cm, 10 cm, 15 cm, 20 cm) while the participants' startle response to each ring was measured. This was repeated for 10 trials.

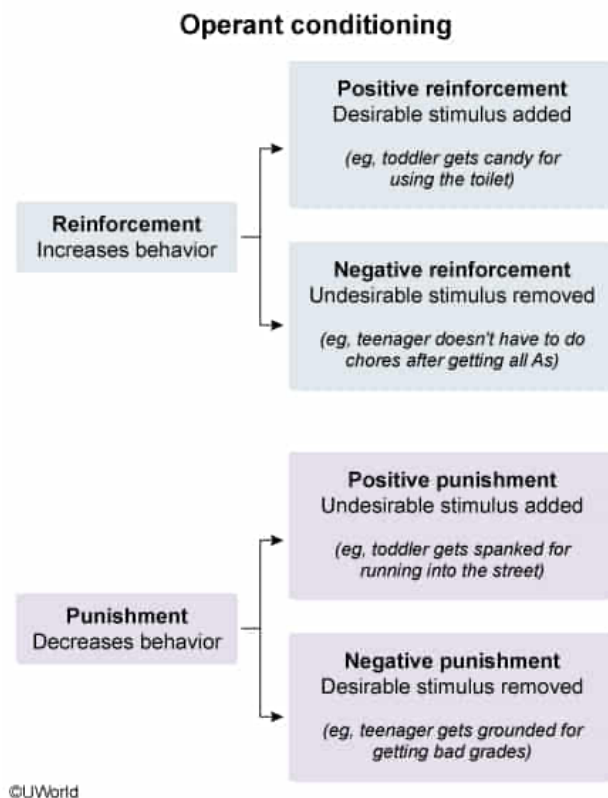
Before Sarah sought therapy, her cleaning rituals acted as:

- ☐ A. positive reinforcement. [36%]
- ✔ ☒ B. negative reinforcement. [41%]
- ☐ C. positive punishment. [19%]
- ☐ D. negative punishment. [2%]

Correct

40% answered correctly

Explanation:



Operant conditioning is a type of learning that occurs when the likelihood of an organism repeating a **behavior** is influenced by the **outcome** of that behavior (ie, reward or punishment). For example, when a rat receives a food pellet (ie, reward) after pushing a lever, the rat is more likely to push the lever again.

Reinforcement increases (or maintains) behavior; **negative reinforcement** occurs when an **undesirable stimulus is removed**, which encourages a behavior to happen again.

Sarah's cleaning rituals act as *negative reinforcement* because she experienced a reduction in anxiety (ie, removal of an undesirable stimulus) after performing cleaning rituals, leading her to perform the cleaning rituals more often (ie, behavior increases).

(Choice A) Positive reinforcement occurs when a *desirable stimulus is applied*, which encourages a behavior to happen again. For Sarah, cleaning rituals remove an unpleasant stimulus (ie, reduce anxiety), an example of negative reinforcement, rather than apply a desirable stimulus.

(Choices C and D) Punishment discourages a behavior from happening again. Positive punishment occurs when an *undesirable stimulus is applied*, and negative

punishment occurs when a *desirable stimulus is removed*. For Sarah, the alleviation of anxiety (ie, removal of unpleasant stimulus) encourages her to perform the cleaning rituals again, which is an example of negative reinforcement, not punishment.

Educational objective:

In operant conditioning, reinforcement increases the likelihood a behavior will occur again. Positive reinforcement involves the application of a desirable stimulus, whereas negative reinforcement involves the withdrawal of an undesirable stimulus.

Time spent:0 sec
Subject:
Behavioral Sciences

QID:402013
System:
Learning, Memory, and Cognition

On the way home from a routine visit to the pediatrician, Sarah stops with her infant son at the local convenience store. As she is waiting in line, a man squeezes in closely behind them and sneezes. As Sarah walks home, she finds that her heart is racing, she feels nauseous, and she is terrified that her child will become severely ill.

Soon after, she becomes obsessed with disease and contamination. She begins to take extensive measures to "decontaminate" herself and her home, including washing her hands dozens of times daily. These cleaning rituals have the immediate effect of lessening Sarah's anxiety, so she soon spends most of her day performing them. She seeks treatment from a therapist, who instructs her to gradually expose herself to "contamination" but refrain from her cleaning rituals. The therapist also teaches Sarah how to recognize and modify her maladaptive thoughts about illness prevention. Within several weeks, Sarah finds that she can manage her thoughts about disease and contamination without resorting to ritualistic cleaning.

A researcher hypothesized that individuals with obsessive-compulsive disorder (OCD), such as Sarah, would show a more generalized fear response to safe stimuli compared to controls. Generalized conditioned fear responses occur when individuals react with fear to stimuli that resemble the original conditioned stimulus (eg, someone injured in an accident with a red truck fears all red vehicles). The researcher conditioned 100 individuals (50 with OCD and 50 controls) to associate a shock with a visual stimulus. In phase 1, participants viewed a computer screen that alternated between images of a small ring (1 cm) and a large ring (20 cm). The small ring and large ring were displayed for 1 second each, with a blank screen in between for 5 seconds. A mild electric shock was paired with the large ring, and no shock was paired with the small ring. This was repeated for 10 trials. In phase 2, shocks were discontinued, and three intermediate-sized rings were added to the series and presented in increasing size (ie, 1 cm, 5 cm, 10 cm, 15 cm, 20 cm) while the participants' startle response to each ring was measured. This was repeated for 10 trials.

The therapeutic techniques described in the second paragraph are most closely related to which approach?

- ☐ A. Psychodynamic [6%]
- ☐ B. Biomedical
- ☐ C. Humanistic [2%]
- ✔ ☒ D. Cognitive-behavioral [90%]

Correct

90% answered correctly

Explanation:

Major psychotherapeutic approaches		
Cognitive-behavioral therapy	Psychoanalytic therapy ("talk therapy")	Humanistic therapy (person-centered therapy)
Attempts to change negative thoughts/beliefs (cognitions) & maladaptive behaviors	Attempts to uncover how unconscious conflicts rooted in childhood shape behaviors	Attempts to empower individual to move toward self-actualization
Various techniques (desensitization, self-talk) used to replace destructive thoughts/behaviors with healthy ones	Various techniques (eg, free association, dream analysis) used to analyze unconscious	Unconditional positive regard & empathy used to encourage client to reach full potential

Cognitive-behavioral therapy (CBT) is a major therapeutic approach designed to help individuals **replace negative thoughts** and **behaviors** with **healthier** thoughts and behaviors.

For example, a cognitive-behavioral therapist treating an individual with social anxiety could focus on replacing negative thoughts (eg, "I'm awkward in social situations") and avoidant behaviors (eg, eating alone) with more positive thoughts (eg, "I'm interesting and a great listener") and sociable behaviors (eg, eating with others).

Sarah's therapist encourages her to expose herself to "contamination" and not engage in cleaning rituals in an attempt to change her maladaptive patterns of behavior. Sarah's therapist also teaches her to recognize and modify her maladaptive thoughts about illness prevention. These

techniques are most closely related to a *cognitive-behavioral* approach.

(Choice A) The psychodynamic approach focuses on revealing the unconscious and understanding how previous conflict influences the present. Replacing negative thoughts and behaviors is most characteristic of a CBT (not psychodynamic) approach.

(Choice B) The **biomedical approach** suggests that physiological causes (eg, abnormal brain chemistry) result in psychological symptoms, and therefore medical interventions (eg, medication) are used as treatment. Biomedical therapy does not focus on replacing negative thoughts and behaviors with healthier ones, as CBT does.

(Choice C) The humanistic approach is holistic and emphasizes higher aspects of human nature, such as the drive toward self-actualization (ie, achieving one's full potential). Replacing negative thoughts and behaviors is more directly characteristic of a CBT (not humanistic) approach.

Educational objective:

Cognitive-behavioral therapy is a major therapeutic approach that attempts to replace negative thoughts and behaviors with healthier thoughts and behaviors.

Time spent:0 sec
Subject:
Behavioral Sciences

QID:402014
System:
Learning, Memory, and Cognition

On the way home from a routine visit to the pediatrician, Sarah stops with her infant son at the local convenience store. As she is waiting in line, a man squeezes in closely behind them and sneezes. As Sarah walks home, she finds that her heart is racing, she feels nauseous, and she is terrified that her child will become severely ill.

Soon after, she becomes obsessed with disease and contamination. She begins to take extensive measures to "decontaminate" herself and her home, including washing her hands dozens of times daily. These cleaning rituals have the immediate effect of lessening Sarah's anxiety, so she soon spends most of her day performing them. She seeks treatment from a therapist, who instructs her to gradually expose herself to "contamination" but refrain from her cleaning rituals. The therapist also teaches Sarah how to recognize and modify her maladaptive thoughts about illness prevention. Within several weeks, Sarah finds that she can manage her thoughts about disease and contamination without resorting to ritualistic cleaning.

A researcher hypothesized that individuals with obsessive-compulsive disorder (OCD), such as Sarah, would show a more generalized fear response to safe stimuli compared to controls. Generalized conditioned fear responses occur when individuals react with fear to stimuli that resemble the original conditioned stimulus (eg, someone injured in an accident with a red truck fears all red vehicles). The researcher conditioned 100 individuals (50 with OCD and 50 controls) to associate a shock with a visual stimulus. In phase 1, participants viewed a computer screen that alternated between images of a small ring (1 cm) and a large ring (20 cm). The small ring and large ring were displayed for 1 second each, with a blank screen in between for 5 seconds. A mild electric shock was paired with the large ring, and no shock was paired with the small ring. This was repeated for 10 trials. In phase 2, shocks were discontinued, and three intermediate-sized rings were added to the series and presented in increasing size (ie, 1 cm, 5 cm, 10 cm, 15 cm, 20 cm) while the participants' startle response to each ring was measured. This was repeated for 10 trials.

What are the conditioned and unconditioned stimuli in the study, respectively?

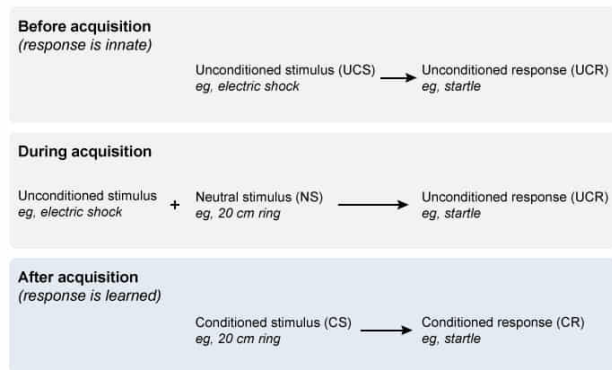
- ☐ A. The 1-cm ring and the electric shock [1%]
- ✓ ☒ B. The 20-cm ring and the electric shock [57%]
- ☐ C. The electric shock and the 20-cm ring [21%]
- ☐ D. The 20-cm ring and the 1-cm ring [19%]

Correct

57% answered correctly

Explanation:

Associative learning: classical conditioning



Classical conditioning occurs when a stimulus that did not previously elicit a meaningful response takes on the properties of a biologically arousing stimulus. For example, when a dog hears the sound of a bell every time it receives food, the sound of the bell alone will eventually produce salivation (eg, a meaningful response).

Unconditioned (ie, biologically arousing) stimuli cause **unconditioned responses** (ie, innate reactions). Neutral stimuli do not produce meaningful responses. After being paired with an unconditioned stimulus, the previously neutral stimulus becomes a **conditioned stimulus** when it causes the organism to react with a learned (ie, conditioned) response, which is typically similar to the unconditioned response.

In this study, the 20-cm ring (neutral stimulus) was paired with an electric shock (unconditioned stimulus), which elicited a startle (unconditioned response). After repeated pairings, the 20-cm ring (originally the neutral stimulus) became the conditioned stimulus because it was associated with the electric shock. After learning has taken place, the 20-cm ring (conditioned stimulus) will provoke a startle (conditioned response).

Therefore, in this study the 20-cm ring was the *conditioned stimulus*, and the electric shock was the *unconditioned stimulus*, not vice versa (**Choice C**).

(**Choices A and D**) The 1-cm ring, which was not paired with the electric shock, remained a neutral stimulus in this study.

Educational objective:

Unconditioned stimuli (eg, food) are biologically arousing, causing an innate

(untrained) reaction called an unconditioned response (eg, salivation). Neutral stimuli (eg, bell) do not produce a meaningful response. After being paired with the unconditioned stimulus, the previously neutral stimulus, now termed the conditioned stimulus, causes the organism to react with a learned, or conditioned, response (eg, salivation in response to a bell).

Time spent:0 sec

Subject:

Behavioral Sciences

QID:402015

System:

Learning, Memory, and Cognition

On the way home from a routine visit to the pediatrician, Sarah stops with her infant son at the local convenience store. As she is waiting in line, a man squeezes in closely behind them and sneezes. As Sarah walks home, she finds that her heart is racing, she feels nauseous, and she is terrified that her child will become severely ill.

Soon after, she becomes obsessed with disease and contamination. She begins to take extensive measures to "decontaminate" herself and her home, including washing her hands dozens of times daily. These cleaning rituals have the immediate effect of lessening Sarah's anxiety, so she soon spends most of her day performing them. She seeks treatment from a therapist, who instructs her to gradually expose herself to "contamination" but refrain from her cleaning rituals. The therapist also teaches Sarah how to recognize and modify her maladaptive thoughts about illness prevention. Within several weeks, Sarah finds that she can manage her thoughts about disease and contamination without resorting to ritualistic cleaning.

A researcher hypothesized that individuals with obsessive-compulsive disorder (OCD), such as Sarah, would show a more generalized fear response to safe stimuli compared to controls. Generalized conditioned fear responses occur when individuals react with fear to stimuli that resemble the original conditioned stimulus (eg, someone injured in an accident with a red truck fears all red vehicles). The researcher conditioned 100 individuals (50 with OCD and 50 controls) to associate a shock with a visual stimulus. In phase 1, participants viewed a computer screen that alternated between images of a small ring (1 cm) and a large ring (20 cm). The small ring and large ring were displayed for 1 second each, with a blank screen in between for 5 seconds. A mild electric shock was paired with the large ring, and no shock was paired with the small ring. This was repeated for 10 trials. In phase 2, shocks were discontinued, and three intermediate-sized rings were added to the series and presented in increasing size (ie, 1 cm, 5 cm, 10 cm, 15 cm, 20 cm) while the participants' startle response to each ring was measured. This was repeated for 10 trials.

A researcher replicates the study but continues to occasionally pair the 20-cm ring with an electric shock in phase 2. Which phenomenon is the

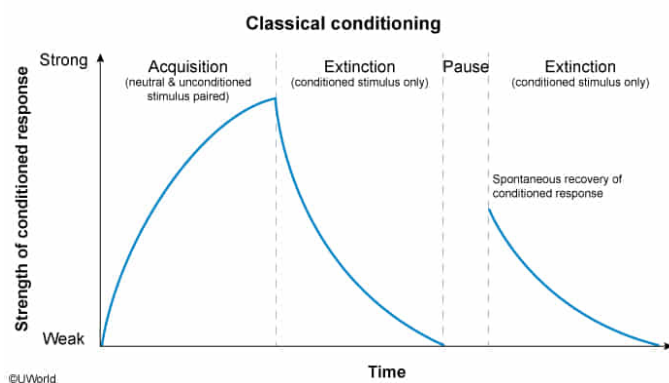
researcher most likely trying to avoid?

- ✓ ☒ A. Extinction [84%]
☐ B. Acquisition [3%]
☐ C. Spontaneous recovery [6%]
☐ D. Stimulus discrimination [5%]

Correct

84% answered correctly

Explanation:



In **classical conditioning**, the strength of the conditioned response tends to follow a characteristic pattern over time:

- **Acquisition** refers to the learning that takes place as an association is formed between the unconditioned stimulus (eg, food) and the neutral stimulus (eg, bell).
- **Extinction** occurs when a conditioned response gradually stops occurring in the absence of the unconditioned stimulus. For example, a dog that has been conditioned to salivate in response to the sound of a bell will gradually stop salivating if the bell is repeatedly presented without food.
- **Spontaneous recovery** occurs when an extinct response reappears after a period of time (eg, a dog will salivate again upon hearing a bell a few days after extinction has occurred).

If the 20-cm ring (conditioned stimulus) was repeatedly presented in the absence of the electric shock (unconditioned stimulus), this would eventually extinguish the startle (conditioned response). However, if an electric shock were occasionally paired with the 20-cm ring, this would maintain the

conditioned fear response, thereby *avoiding extinction*.

(Choice B) Phase 1 of the study, when the electric shock was repeatedly paired with the 20-cm ring, was the acquisition phase. Occasionally administering the electric shock with the 20-cm ring during phase 2 would not *avoid acquisition*, which has already occurred.

(Choice C) Spontaneous recovery is the reappearance of the conditioned response after it has already been extinguished. Occasionally administering the electric shock during phase 2 would not *avoid spontaneous recovery*, which only occurs after a conditioned response has become extinct.

(Choice D) Stimulus discrimination occurs when an organism distinguishes between two similar stimuli (eg, a dog that has been conditioned to salivate in response to a bell salivates only to the sound of *that* bell, not in response to other similar sounds). This concept does not apply to administering the electric shock during phase 2.

Educational objective:

In classical conditioning, extinction occurs when the conditioned stimulus alone no longer produces the conditioned response.

Time spent:0 sec
Subject:
Behavioral Sciences

QID:402016
System:
Learning, Memory, and Cognition

On the way home from a routine visit to the pediatrician, Sarah stops with her infant son at the local convenience store. As she is waiting in line, a man squeezes in closely behind them and sneezes. As Sarah walks home, she finds that her heart is racing, she feels nauseous, and she is terrified that her child will become severely ill.

Soon after, she becomes obsessed with disease and contamination. She begins to take extensive measures to "decontaminate" herself and her home, including washing her hands dozens of times daily. These cleaning rituals have the immediate effect of lessening Sarah's anxiety, so she soon spends most of her day performing them. She seeks treatment from a therapist, who instructs her to gradually expose herself to "contamination" but refrain from her cleaning rituals. The therapist also teaches Sarah how to recognize and modify her maladaptive thoughts about illness prevention. Within several weeks, Sarah finds that she can manage her thoughts about disease and contamination without resorting to ritualistic cleaning.

A researcher hypothesized that individuals with obsessive-compulsive disorder (OCD), such as Sarah, would show a more generalized fear response to safe stimuli compared to controls. Generalized conditioned fear responses occur when individuals react with fear to stimuli that resemble the original conditioned stimulus (eg, someone injured in an accident with a red truck fears all red vehicles). The researcher conditioned 100 individuals (50 with OCD and 50 controls) to associate a shock with a visual stimulus. In phase 1, participants viewed a computer screen that alternated between images of a small ring (1 cm) and a large ring (20 cm). The small ring and large ring were displayed for 1 second each, with a blank screen in between for 5 seconds. A mild electric shock was paired with the large ring, and no shock was paired with the small ring. This was repeated for 10 trials. In phase 2, shocks were discontinued, and three intermediate-sized rings were added to the series and presented in increasing size (ie, 1 cm, 5 cm, 10 cm, 15 cm, 20 cm) while the participants' startle response to each ring was measured. This was repeated for 10 trials.

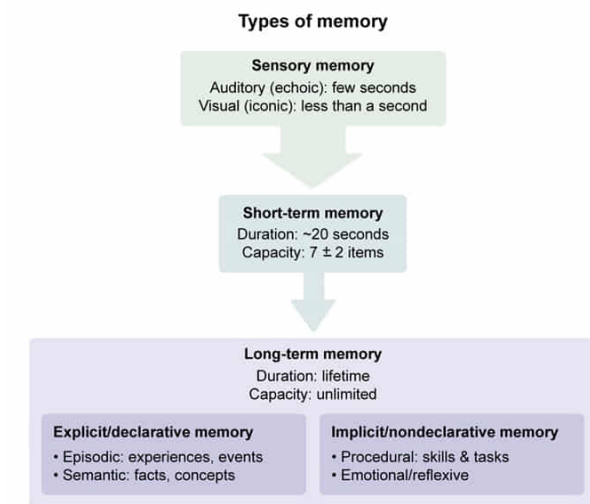
A conditioned fear response during phase 2 occurs as a result of which type of memory?

- ✔ ☒ A. Implicit memory [50%]
- ☐ B. Procedural memory [12%]
- ☐ C. Explicit memory [24%]
- ☐ D. Semantic memory [12%]

Correct

50% answered correctly

Explanation:



©UWorld

Research suggests that humans have two long-term memory systems, referred to as implicit and explicit. **Implicit/nondeclarative memory** is memory for things that **cannot be consciously recalled**, such as skills, tasks, emotions, and reflexes.

Examples of implicit memory include:

- Procedural memory, which is memory for motor skills (eg, riding a bicycle) **(Choice B)**.
- Emotional/reflexive memory, which is memory for associations between stimuli (eg, salty ocean air triggers pleasant emotions from childhood beach vacations).

Classically conditioned fear responses are emotional, outside of conscious recall (ie, automatic/involuntary), and often elicit reflexive responses (eg, startle reflex). Therefore, the conditioned fear response during phase 2 occurs as a result of *implicit memory*.

(Choices C and D) Explicit/declarative memory is memory for facts and events that can be consciously or intentionally recalled. Explicit memory consists of episodic memory, which is memory for personal experiences (eg, what you ate for dinner last night), and semantic memory, which includes knowledge about facts (eg, apples are a type of fruit).

Educational objective:

Implicit/nondeclarative memory is memory for things that cannot be consciously recalled, such as skills, tasks, emotions, and reflexes. Classically conditioned responses rely on implicit memory.

Time spent:0 sec
Subject:
Behavioral Sciences

QID:402017
System:
Learning, Memory, and Cognition

On the way home from a routine visit to the pediatrician, Sarah stops with her infant son at the local convenience store. As she is waiting in line, a man squeezes in closely behind them and sneezes. As Sarah walks home, she finds that her heart is racing, she feels nauseous, and she is terrified that her child will become severely ill.

Soon after, she becomes obsessed with disease and contamination. She begins to take extensive measures to "decontaminate" herself and her home, including washing her hands dozens of times daily. These cleaning rituals have the immediate effect of lessening Sarah's anxiety, so she soon spends most of her day performing them. She seeks treatment from a therapist, who instructs her to gradually expose herself to "contamination" but refrain from her cleaning rituals. The therapist also teaches Sarah how to recognize and modify her maladaptive thoughts about illness prevention. Within several weeks, Sarah finds that she can manage her thoughts about disease and contamination without resorting to ritualistic cleaning.

A researcher hypothesized that individuals with obsessive-compulsive disorder (OCD), such as Sarah, would show a more generalized fear response to safe stimuli compared to controls. Generalized conditioned fear responses occur when individuals react with fear to stimuli that resemble the original conditioned stimulus (eg, someone injured in an accident with a red truck fears all red vehicles). The researcher conditioned 100 individuals (50 with OCD and 50 controls) to associate a shock with a visual stimulus. In phase 1, participants viewed a computer screen that alternated between images of a small ring (1 cm) and a large ring (20 cm). The small ring and large ring were displayed for 1 second each, with a blank screen in between for 5 seconds. A mild electric shock was paired with the large ring, and no shock was paired with the small ring. This was repeated for 10 trials. In phase 2, shocks were discontinued, and three intermediate-sized rings were added to the series and presented in increasing size (ie, 1 cm, 5 cm, 10 cm, 15 cm, 20 cm) while the participants' startle response to each ring was measured. This was repeated for 10 trials.

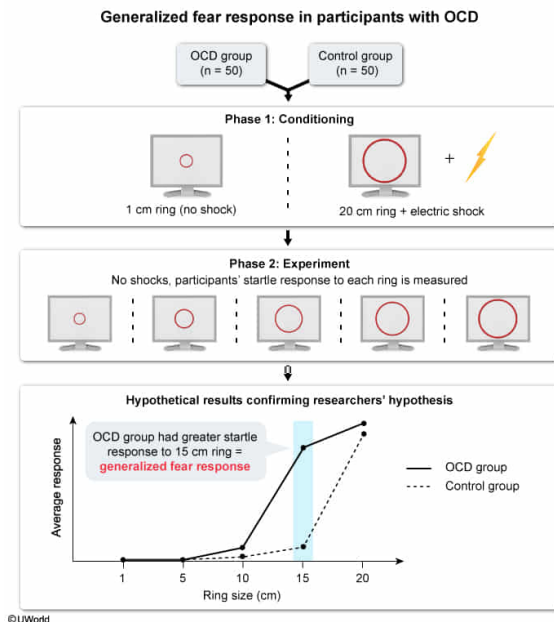
Which hypothetical result would provide support for the researcher's hypothesis?

- ☐ A. Compared to participants with OCD, controls show a greater startle response to the 1-cm ring. [4%]
- ☐ B. Controls and participants with OCD do not differ in startle response to any of the rings. [1%]
- ☒ C. Compared to controls, participants with OCD show a greater startle response to the 15-cm ring. [86%]
- ☐ D. Controls and participants with OCD show the greatest startle response to the 20-cm ring. [7%]

Correct

86% answered correctly

Explanation:



Stimulus generalization occurs when the conditioned response is elicited by stimuli **similar to the original conditioned stimulus**. For example, an infant who has been conditioned to fear a white rat may also display fear in response to other stimuli, such as a white rabbit or white ball of yarn.

In this study, all participants were conditioned to associate the 20-cm ring with an electric shock. After conditioning, the presentation of the 20-cm ring (conditioned stimulus) would elicit a startle (conditioned response).

The researcher hypothesized that compared to controls, participants with OCD would show a *more generalized* fear response to safe stimuli. Therefore, a finding that participants with OCD showed a greater startle response to the 15-cm ring (ie, a safe stimulus that resembles the conditioned stimulus) than controls supports the hypothesis.

(Choice A) The finding that controls startle more to a safe stimulus (ie, 1-cm ring) than the participants with OCD does not support the researcher's hypothesis (ie, it is the opposite finding).

(Choice B) Finding no difference between the groups does not support the researcher's hypothesis that the participants with OCD would show a more generalized fear response to safe stimuli.

(Choice D) Both groups would be expected to show the greatest startle response to the 20-cm ring; however, the researcher's hypothesis did not address the intensity of startle response to the 20-cm ring relative to the other rings. Therefore, both groups showing the greatest startle response to the 20-cm ring would not support the hypothesis.

Educational objective:

Stimulus generalization occurs when stimuli (eg, white rabbit) similar to the original conditioned stimulus (eg, white rat) elicits the conditioned response (eg, fear).